Muhammad Shahid

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Education

University of Alberta **2025 – Present**

- Bachelor of Science, Computer Science
- Coursework: OOP, Data Structures & Algorithms, Machine Learning, Software Engineering

Projects

PocketPal – Expense Tracker | *Python, Flask, SQLite, Jinja2, Chart.js, REST APIs, OAuth 2.0*

- Built a full-stack personal finance web application with 2 secure login paths: email/password and Google OAuth 2.0.
- Engineered CRUD expense management with multi-field filtering (date range, category, amount) through RESTful Flask routes.
- Created an interactive Chart.js dashboard that turns raw transactions into monthly spending trends and category-level breakdowns at a glance.
- Automated expense entry with an OCR receipt scanner that pre-fills the form and flags likely duplicates (same date, amount within 5%).
- Added one-click CSV and PDF export, plus monthly budget limits with a live progress bar that shows spending against the limit.
- Designed a normalized SQLite schema with foreign keys and server-side validation, with password complexity rules and encrypted credential storage.

Ledger – Job Application Tracker | *Python, Flask, PostgreSQL, Supabase, JavaScript, Jinja2*

- Built a drag-and-drop Kanban board across 4 pipeline stages (Applied, Interview, Offer, Rejected) with instant card moves and real-time status persistence via RESTful Flask routes.
- Migrated the data layer from SQLite to Supabase-hosted PostgreSQL using incremental schema migrations, preserving all existing user data.
- Engineered authentication combining Supabase Auth and Google OAuth 2.0, supporting login with either username or email.
- Developed deadline tracking with dynamic urgency indicators so upcoming interviews stand out, plus soft-delete archiving with restore and one-click CSV export.
- Implemented a responsive, mobile-first interface with a swipeable touch-friendly board and persistent dark mode.

AI Spending Assistant | *React, Flask, Python, LLM API, REST APIs*

- Built a React chat interface that answers natural-language questions about a user's live expense data through an LLM API.
- Engineered the backend to precompute every financial calculation (totals, category breakdowns, monthly averages) in Python before the LLM sees it, cutting total-spending error from $107 to $0 and matching all 8 category totals exactly.
- Replaced session cookies with signed, expiring tokens, fixing cross-origin login failures caused by third-party cookie restrictions across devices and browsers.
- Built a RESTful Flask endpoint that queries and categorizes a user's live expense data on every request, supporting accurate follow-up questions.

Skills

Languages: Python, C, JavaScript, HTML/CSS, SQL
AI & ML: LLM API integration, Microsoft Copilot Studio
Frameworks & Web: React, Flask, Jinja2, Chart.js
Data & Cloud: PostgreSQL, SQLite, Supabase, Render, Pandas, NumPy, Matplotlib
Tools & Methods: GitHub, Linux, VS Code, REST APIs, OAuth 2.0, Token-Based Auth
Productivity: Google Workspace, Microsoft Office 365